Multilevel models for longitudinal data

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Motivation

Longitudinal studies are very common in various fields like psychology, medicine, education, economics, and sociology, and numerous examples and collections of longitudinal data can be found online (see, for instance, Atlas of longitudinal studies). Longitudinal studies are designed to observe a sample over time to extract knowledge about patterns of trends and fluctuations in the general population. The same group of subjects, which may be single persons or groups/organizations, is tracked over time and its characteristics of interest are observed.

In this work, we will take a closer look at an alternative method to analyze longitudinal data using the multilevel modeling framework. This method has several advantages over the traditional approaches and allows analyzing more real-world longitudinal datasets. Here we will use an example dataset to showcase the multilevel modeling approach and go through the analysis step by step, providing R code to build models and create figures and its output.

Setup

Import

In the following we will use implementation of multilevel models from the package `lme4` [1] as well as statistical tests from package `lmerTest` [3]. We mostly use `ggplot2` [5] for graphics and `knitr` and `pander` [6] for R Markdown generation.

```
library("lme4")
library("lmerTest")
library("ggplot2")
library("dplyr")
```

Example dataset

As mentioned above we will use an example to illustrate multilevel approach. Here a dataset from the collection of Centre for Multilevel Modelling, University of Bristol is taken, which can be accessed via this link. We have chosen a dataset describing weight gains of 568 Asian children in a British community over several weeks after birth. The children were measured on up to five occasions visiting a clinic roughly to target examination dates of 6 weeks, then 8, 12, 27 months. Additional variables recorded were weight at birth and sex of a baby.

The code below shows how to load the dataset and convert it to format which suits best for further analysis.

```
# Read the file line by line and convert to one single string
data_path = "asian/ASIAN.DAT"
data_raw <- toString(readLines(data_path))
```

```r
# Split the long string by commas
data_raw <- strsplit(data_raw, ", ")

# Create a one-column data frame
dataset <- data.frame(data_raw)
colnames(dataset) <- "raw"

# Extract columns
extract_columns <- function(idxs, df){
    column = sapply(df, substr, start = idxs[1], stop = idxs[2])
    return(column)}

# Define start and end positions for each column
col_idxs = list(c(1,4), # ChildID: characters 1-4
                c(5,7), # Age: characters 5-7
                c(8,12), # Weight: characters 8-12
                c(13,16), # Birthweight: characters 13-16
                c(17,17)) # Gender: character 17

col_list <- lapply(col_idxs, extract_columns, df = dataset) # 5

# Assign extracted columns to new variables in the data frame
dataset$ChildID <- as.vector(col_list[[1]])
dataset$Age <- as.vector(col_list[[2]])
dataset$Weight <- as.vector(col_list[[3]])
dataset$Birthweight <- as.vector(col_list[[4]])
dataset$Gender <- as.vector(col_list[[5]])

dataset <- dataset[,-1]
```

Challenges of longitudinal data

Levels of analysis and time continuum

Longitudinal studies are very popular despite being quite complex and expensive to convey [2]. It naturally depends on the scope of a study: how large the study sample is, how long are study subjects tracked for and even how probable a drop-out is. However, people are hard to track in general and measures need to be well-documented over time and comparable, i.e. measure methods need to stay consistent.

Nevertheless, analysis of repeated measures of several subjects is quite useful. They make different levels of analysis accessible:

1. Within-person / intra-individual analysis: relationships among intra-individual differences in time-variant variables. (level 1 analysis)

2. Between-person / inter-individual analysis: relationships among inter-individual differences in time-invariant variables. (level 2 analysis)

It is also important to differentiate, whether change or fluctuation in target value should be investigated. In Hoffman [2] the two terms are use to describe different kinds of longitudinal variability. Here, change describes systematic increase or decrease over time, often in longer perspective, while fluctuation usually refers to undirected variation over time on short-term horizons.

Model for mean and model for variance

A model for mean describes structural part of the model, i.e. how does the outcome depend on regressor values for a study subject. It includes *fixed effects* of the covariates, where *fixed* means that the effect is same for all subjects. In the model for the mean a choice should be made, what regressors should be taken in account and in which form. This choice is usually based on specific research question.

A model for variance describes, on the other hand, the distribution of residuals of the outcome. Separate model for variance allows us to model patterns of variance and covariance in residuals, which introduces great flexibility compared to overly simplifying assumptions for variance structure in generalized linear models or ANOVA. There residuals are often assumed to be normally distributed and independent across persons, with constant variance across persons. That often doesn't hold for real-world longitudinal data [2]. Modeling variance separately also allows to include so-called *random effects* of covariates, when every study subject has its own effect of the same predictor. Addition of random effects allow greater flexibility - variance between and within a person can be distinguished.

Models for repeated measures

Traditional approach for modeling longitudinal data - **R**epeated **M**easures **An**alysis **o**f **Va**riance (RM-ANOVA) accounts for dependency of residuals within one unit/subject, but still assumes normal distribution of residuals as well as homoscedasticity (constant variance over time points) and compound symmetry (equal variances and equal covariances over subjects).

Moreover, the goal of RM-ANOVA model is not to represent the change and fluctuation in the target variable in some general functional form, but to "reproduce" the observed mean per occasion using fixed effects equal to the number of occasions minus 1 [2, p.96]. This sets an additional constraint to data such as equal intervals between observation time points as well as same time points for every participant, which is often hard to implement in real life.

Also one could imagine that it may happen quite often, that a participant doesn't show up once or drops of out of study completely due to any possible reason. RM-ANOVA cannot handle missing data, because it requires all observations for all subjects and time points to be present. Therefore all those subjects where at least one observation is missing is removed from the dataset. And many datasets cannot be analyzed with this method at all. Multilevel models for longitudinal data cover all of the named disadvantages and have less strict assumptions about data distribution. Therefore, they represent an attractive alternative to RM-ANOVA.

Data preprocessing

An important part of any analysis is data exploration and preparation. In the example above, the dataset consists of five variables: ID of a child, age (in days), weight (in grams), birth weight (in grams) and gender (encoded as 1 or 2). First of all, we want to ensure the correct type of variable for each column (integer/factor/string). We also change encoding of gender to dummy encoding (with 1 for girls and 0 for boys respectively) and store numeric values and category name in two separate columns (see code below).

We also calculate number of observations for each child to see, whether the dataset is balanced across subjects. We can already see from the first 6 observations that this is not the case here.

```
# Preprocess columns
dataset[, 2:4] <- sapply(dataset[,2:4], as.integer)
dataset$ChildID <- as.factor(gsub(" ", "", dataset$ChildID))

# Decode gender
dataset$GenderID <- as.integer(dataset$Gender) - 1
dataset$Gender <- factor(dataset$GenderID, labels = c("Boy", "Girl"))

# Number of observations
dataset$NObs <- as.vector(table(dataset$ChildID)[dataset$Child])
kable(head(dataset))
```

ChildID	Age	Weight	Birthweight	Gender	GenderID	NObs
2	377	10262	3360	Girl	1	2
2	825	14500	3360	Girl	1	2
3	54	4053	3310	Girl	1	4
3	242	8136	3310	Girl	1	4
3	377	9752	3310	Girl	1	4
3	825	13180	3310	Girl	1	4

Furthermore, we can plot observations against time to see, how much the measurement times vary from child to child - we already know from the dataset description, that intervals between measurements are not equal and may differ from case to case (see Figure 1).

```
plot_all_obs <- function(varx, vary, df,
                         transfx = "", transfy = ""){
    ggplot(df, aes(x = .data[[varx]], y = .data[[vary]])) +
    coord_cartesian(xlim = c(min(df[varx]), max(df[varx])),
                    ylim = c(min(df[vary]), max(df[vary]))) +
    geom_line(aes(group=ChildID), col = "darkgray") +
    geom_point(aes(fill=as.factor(ChildID)), pch=21, size=2, stroke=1) +
```

```
    scale_x_continuous(name = paste0("Age",transfx),
                       breaks=round(seq(min(df[,varx]), max(df[,varx]),
                                        length.out = 10)))  +
    scale_y_continuous(name = paste0("Weight", transfy),
                       limits=range(df[,vary])) +
    theme_bw() + theme(axis.text.x=element_text(size=12, colour="black"),
                       axis.text.y=element_text(size=12, colour="black"),
                       axis.title=element_text(size=12,face="bold")) +
    theme(strip.text.x = element_text(size = 12)) +
    theme(legend.position="none")}

plot_all_obs("Age", "Weight", dataset,
             transfx = " (in days)", transfy = " (in grams)")
```

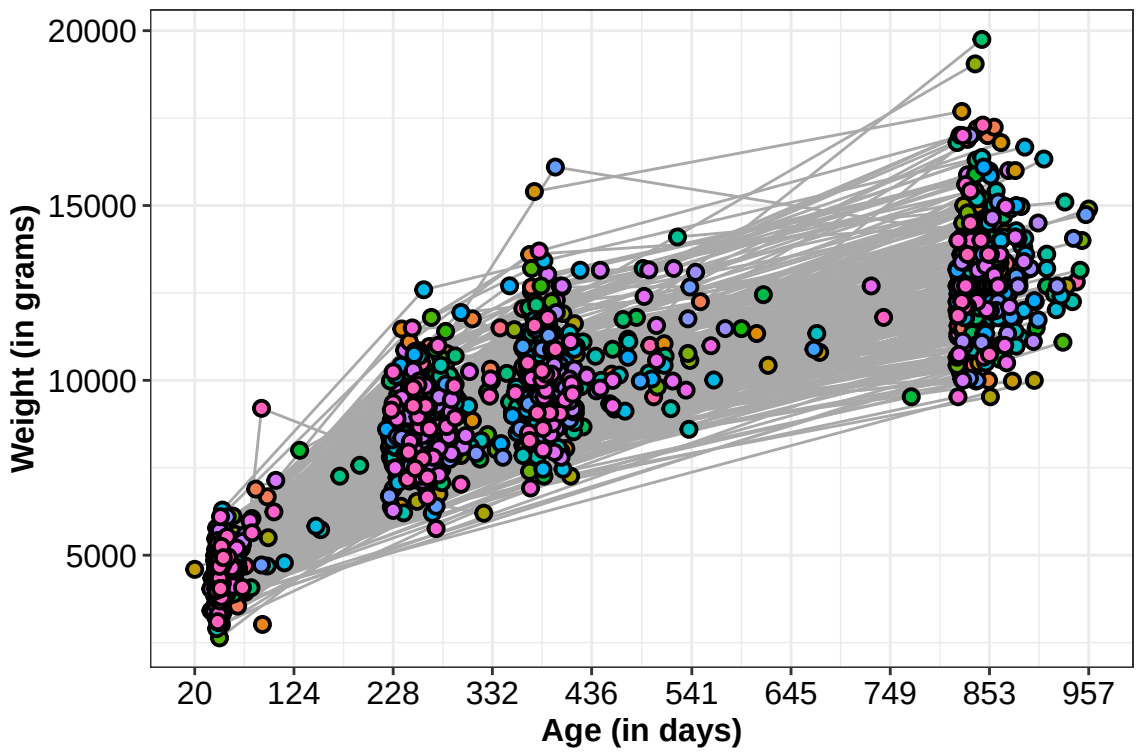

Figure 1: Children weight observations over time

To observe data imbalance we can create multiple plots, grouping data by number of observations for single child (see Figure 2). This can be done by adding a line `facet_wrap(~.data[["NObs"]])` to the definition of `plot_all_obs()` function above (here we created a separate function with alterations). We can also calculate absolute frequency for number of visits depending on gender as in Table 2.

```r
plot_facet_var("Age", "Weight", df=dataset, "NObs",
               transfx = " (in days)", transfy = " (in grams)")
```

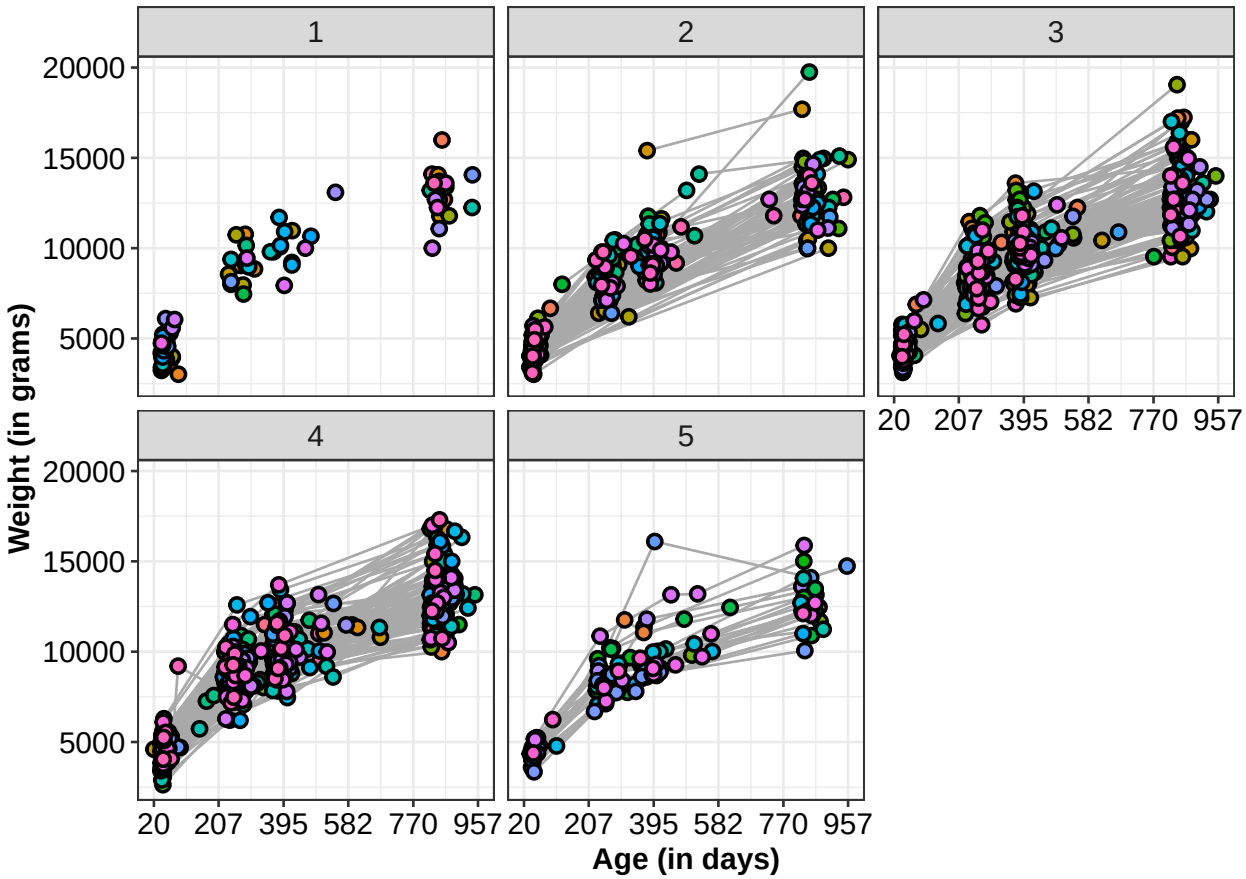

Figure 2: Original observations by number of visits in the clinic

```r
kable(table(dataset$Gender, dataset$NObs) / rep(1:5, each = 2))
```

Table 2: Number of participants per number of observations

	1	2	3	4	5
Boy	45	62	86	75	15
Girl	46	79	83	68	9

Here several observations can be made. First of all, both number or observations and measurement time points vary strongly over participants, which justifies use of multilevel models to analyse this dataset. Secondly, 91 participants were only observed once after birth, which

makes these observations not longitudinal at all. They will be excluded for later. Thirdly, dataset is balanced w.r.t. gender distribution. Lastly, the original scale of the variables produces quite large numerical values, which may negatively affect the numerical estimation of models later. Therefore we need to rescale both target variable and covariates. For this we will first convert grams into kilograms and days into years and then center rescaled variables by subtracting sample mean:

```
# Exclude the data with 1 observation only (not longitudinal data)
dataset <- dataset[dataset$NObs > 1,]

dataset$r_Weight <- dataset$Weight / 1000
dataset$r_Birthweight <- dataset$Birthweight / 1000
dataset$r_Age_years <- dataset$Age / 365

dataset$c_Age_years <- (dataset$r_Age_years - mean(dataset$r_Age_years))
dataset$c_Weight_kg <- (dataset$r_Weight - mean(dataset$r_Weight))
dataset$c_Birthweight_kg <- (dataset$r_Birthweight -
                              mean(dataset$r_Birthweight))

df_final = dataset[c("ChildID", "Gender", "NObs", "GenderID",
                     "c_Weight_kg", "c_Birthweight_kg", "c_Age_years")]

colnames(df_final) <- c("ChildID", "Gender", "NObs", "GenderID",
                        "Weight", "Birthweight", "Age")
```

We can also observe, that intercept and slope varies from participant to participant, which can also be modeled by multilevel models. In addition, growth rate is not constant - babies grow very fast directly after birth and then growth rate reduces after some time, - which means that we can include some non-linearity in the model to improve the fit.

Multilevel models of growing flexibility

We will fit different multilevel models of increasing complexity on this and compare them based on different criteria. Using these models we can try to answer following research questions, for example:

1. How children's weight changes in the early stages of life?
2. Whether and how those growth patterns differ between individuals?
3. Are there any differences in development between children of different gender?
4. Is there any influence of birth weight on post-natal development?

Random intercept model

Multilevel models are sometimes called mixed models, and these two names highlight different aspects of the very same method. They are called "mixed" because both fixed and

random effects of predicted are allowed, and "multilevel" stands for separate modeling of parameters and their (co-)variances, i.e. on different levels. Key idea is that the observed variability comes from two sources: from both between-person differences and within-person fluctuations. All multilevel models, as stated above, assume observations from different subjects to be independent, but correlated within the subject. Here the dependent variable is assumed to be normally distributed.

The simplest multilevel model would be a random intercept model. That means that we have constant predictions for a subject over time, but we allow the intercept to vary over subjects. It can be written as follows in multilevel notation:

Level 1: $y_{ti} = \beta_{0i} + e_{ti}, \qquad e_{ti} \sim \mathcal{N}(0, \sigma_e^2)$

Level 2: $\beta_{0i} = \gamma_{00} + U_{0i}, \qquad U_{0i} \sim \mathcal{N}(0, \tau_{U_0}^2)$, where:

- y_{ti} — observation of dependent variable of unit i at time point $t \in \{t_1, \ldots, t_N\}$;
- γ_{00} — fixed intercept;
- U_{0i} — random intercept, varies over individuals.

This model has 3 parameters to estimate: the fixed intercept γ_{00}, variance of random intercept $\tau_{U_0}^2$ and residual variance σ_e^2.

We can fit a model using `lmer()` function from `lme4` package (loaded above). The random effects are specified in brackets with the grouping variable for the corresponding level behind the vertical bar "|":

```
model01 <- lmer(Weight ~ 1 + (1|ChildID),
                data=df_final, REML=TRUE)
```

Model parameters are estimated using Maximum-Likelihood (ML) approach, and in `lme4` package both unrestricted and restricted version are implemented. Restricted ML (`REML = TRUE`) accounts for uncertainty from simultaneous estimation of fixed parameters and therefore delivers unbiased parameter variance estimates [2].

```
time_grid <- seq(min(df_final["Age"]), max(df_final["Age"]), length.out = 50)

plot_results <- function(data, model, model_name,
                         n = 10, ncols = 5){
  set.seed(42)
  sample_n_id <- sample(as.vector(unique(data$ChildID[order(data$ChildID)])), n)
  sample_n_id_idx <- which(data$ChildID %in% sample_n_id)
  data_n <- data[sample_n_id_idx,
                 c("ChildID", "Birthweight", "GenderID", "Age", "Weight")]
  unique_data_n <- unique(data_n[c("ChildID", "Birthweight", "GenderID")])
  newdata <- cbind(unique_data_n,
                   rep(rownames(unique_data_n), each = length(time_grid)))[,1:3]
```

```
  newdata$Age <- rep(time_grid, each = n)

  newdata$Predictions <- predict(model, newdata)

  g = ggplot(data_n, aes(x = Age, y = Weight)) +
    geom_point(fill="grey", pch=21, size=2, stroke=1.25) +
    geom_line(aes(group=ChildID)) +
    facet_wrap(~ChildID, ncol=ncols)+
    scale_x_continuous(name = "Age") +
    scale_y_continuous(name = "Weight") +
    geom_line(data = newdata,
              aes(x = Age, y = Predictions, group = ChildID,
                  color = factor(ChildID)), lwd=1.5) +
    ggtitle(model_name)

  return(g)}

plot_results(df_final, model01, n=5, ncols=5,
             model_name = "Model 1: random intercept model")
```

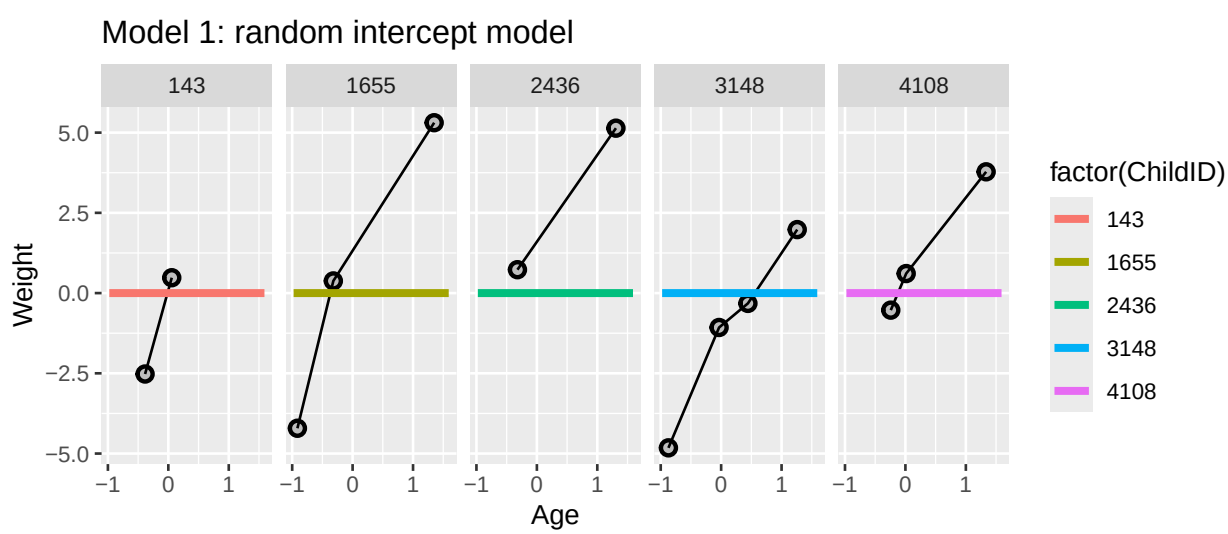

Figure 3: ...

First we observe the model fit on a small subsample of participants. For that we generate multiple time points and use fitted parameter values to plot the model prediction.

In the Figure . . . we can see that the model clearly underfits the observations, which is quite natural: body weight increases while the child grows. Another point is that fit of the model is singular, i.e. variance of random effects is estimated close to 0, which is not very different from just fixed intercept. However, this model serves us later as baseline.

Random intercept, fixed linear time model

We can improve the model fit by adding covariates to the model, and a first candidate would be a fixed linear effect of time γ_{10}. A model can be written as:

Level 1: $y_{ti} = \beta_{0i} + \beta_{1i} \cdot x_{ti} + e_{ti}, \quad e_{ti} \sim \mathcal{N}(0, \sigma_e^2)$

Level 2: $\beta_{0i} = \gamma_{00} + U_{0i}$, where $U_{0i} \sim \mathcal{N}(0, \tau_{U_0}^2), \quad \beta_{1i} = \gamma_{10}$,

where x_{ti} stands for time variable at observation t of individual i.

Hence, only the model for means is changed here and model for variance remains the same. The resulting model will produce growth lines with different intercepts but same slope for all subjects. We can now fit the model and visualize the results:

```
model02 <- lmer(Weight ~ 1 + Age + (1|ChildID),
                    data=df_final, REML=TRUE)

plot_results(df_final, model02, n=5, ncol = 5,
             model_name = "Model 2: random intercept, fixed linear time")
```

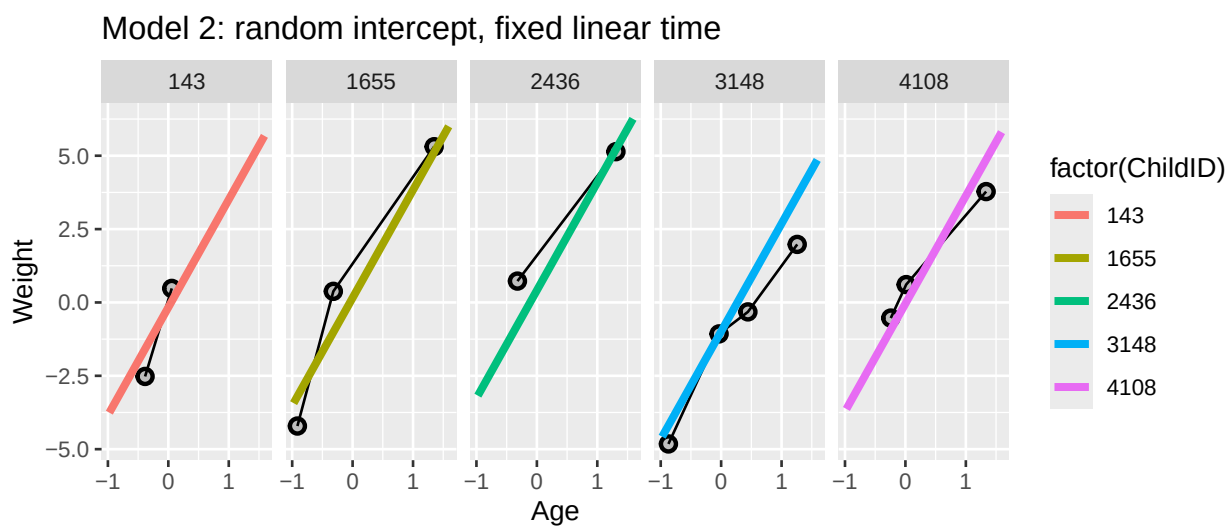

Figure 4: Model predictions with random intercept and fixed slope

Just after visual examination (see Figure 2) we can say that the model fit improved greatly, even though the lines don't overlap with observations perfectly. We can also review model summary for estimation details. First of all, residuals are distributed around zero which may be an indicator for a good fit. Random intercept U_{0i} has estimated variance of $\hat{\tau}_{U_0}^2 = 0.53 \pm 0.73$ and residual variance is estimated at $\hat{\sigma}_e^2 = 1.76 \pm 1.33$.

Linear effect of time is estimated at $\hat{\gamma}_{10} = 3.67$ is considered significant (according to Wald test with $\frac{\hat{\gamma}_j}{SE(\hat{\gamma}_j)} \sim t_{N-f}$, where $j = 1, \ldots, f$ and f is number of fixed effects), however, the fixed intercept ($\hat{\gamma}_{00} = -0.02$) is not significant. Small correlation of fixed effects is also a good sign, meaning that fixed parameters are uncorrelated.

```r
summary(model02)

# Formula: Weight ~ 1 + Age + (1 | ChildID)
#    Data: df_final
#
# REML criterion at convergence: 5358.1
#
# Scaled residuals:
#     Min      1Q  Median      3Q     Max
# -2.3687 -0.7542  0.0667  0.6669  4.4676
#
# Random effects:
#  Groups   Name        Variance Std.Dev.
#  ChildID  (Intercept) 0.5329   0.730
#  Residual             1.7598   1.327
# Number of obs: 1481, groups:  ChildID, 477
#
# Fixed effects:
#               Estimate Std. Error         df t value Pr(>|t|)
# (Intercept)   -0.01564    0.04851  456.71250  -0.322    0.747
# Age            3.67481    0.04495 1139.18833  81.746   <2e-16 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# ----------------------------
# Correlation of Fixed Effects:
#     (Intr)
# Age 0.002
```

Random linear time model

We can go further and add random effect of time into the model. This change will now affect model for the variance:

Level 1: $y_{ti} = \beta_{0i} + \beta_{1i} \cdot x_{ti} + e_{ti}, \quad e_{ti} \sim \mathcal{N}(0, \sigma_e^2)$

Level 2: $\beta_{0i} = \gamma_{00} + U_{0i}, \quad \beta_{1i} = \gamma_{10} + U_{1i},$

where $U_{0i} \sim \mathcal{N}(0, \tau_{U_0}^2)$ and $U_{1i} \sim \mathcal{N}(0, \tau_{U_1}^2)$. This model has 6 parameters to estimate: fixed intercept γ_{00}, fixed slope γ_{10}, residual variance σ_e^2, variances of random effects $\tau_{U_0}^2$ and $\tau_{U_1}^2$ as well as their covariance $\tau_{U_{01}}$.

According to [2] it is often useful to summarize them as covariance matrix G:

$$G = \begin{pmatrix} \tau_{U_0}^2 & \tau_{U_{01}} \\ \tau_{U_{01}} & \tau_{U_1}^2 \end{pmatrix},$$

while residual variance is denoted as matrix R. All variances and covariances are assumed constant over subjects.

We can then reformulate the model equation in vector notation, which will result in

$$Y_i = X_i \cdot \gamma + Z_i \cdot U_i + E_i,$$

where $Y_i \in \mathbb{R}^{n_i}$ is target vector of individual i (at n_i different time points), $U_i = (U_{0i}, U_{0i})^T$ is matrix of random effects and $E_i \in \mathbb{R}^{n_i}$ summarizes the residuals.

Here $E_i \sim \mathcal{N}(0, R)$, $R = \sigma_e^2 I_{n_i}$ is assumed, however, another covariance structures (such as AR(1)) can be explored and assessed.

Two design matrices X_i and Z_i are constructed as follows:

$$X_i = \begin{pmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_{n_i} \end{pmatrix} \in \mathbb{R}^{n_i \times f} \quad Z_i = \begin{pmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_{n_i} \end{pmatrix} \in \mathbb{R}^{n_i \times u},$$

where f is number of fixed effects, and u - number of random effects in the model. Now, using vector notation we can also derive the expression for the total variance (for an individual i):

$$V_i = \mathbb{V}ar(Y_i) = \mathbb{V}ar(X_i \cdot \gamma + Z_i \cdot U_i + E_i) = Z_i^T \, \mathbb{V}ar(U_i) \, Z_i + R = Z_i^T G Z_i + R.$$

Thanks to the construction of the design matrix Z_i, the model can accommodate data from irregular observations with different number of time n_i points per person i. Complete Z matrix is a block-diagonal matrix with matrices $Z_i, i = 1, \ldots, N$ on its diagonal.

Another assumptions of the model is that no covariances are allowed for the variance terms across levels, i.e. between $\tau_{U_0}^2$ and σ_e^2, and $\tau_{U_1}^2$ and σ_e^2 respectively. Also, all units are assumed conditionally independent, meaning no relationships within each of the U_{0i}, U_{1i}, or e_{ti} values across persons are allowed after including the model predictors.

Since we assume residuals to be multivariate normally distributed, likelihood function for the model can be written as follows:

$$L = \prod_{i=1}^{N} (2\pi)^{-\frac{n}{2}} \cdot |V_i|^{-\frac{1}{2}} \cdot \exp\left[-\frac{1}{2}(Y_i - X_i\gamma)^T V_i^{-1} (Y_i - X_i\gamma)\right]$$

with log-likelihood:

$$LL = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{i=1}^{N} \log|V_i| - \frac{1}{2}\sum_{i=1}^{N} (Y_i - X_i\gamma)^T V_i^{-1} (Y_i - X_i\gamma)$$

Newton-Raphson algorithm is then used update estimates of matrix V_i iteratively using derivatives of LL-function. For restricted ML method following function is optimized:

$$LL_{\text{REML}} = -\frac{T-f}{2}\log(2\pi) - \frac{1}{2}\sum_{i=1}^{N} \log|V_i| - \frac{1}{2}\sum_{i=1}^{N} (Y_i - X_i\gamma)^T V_i^{-1} (Y_i - X_i\gamma) - \frac{1}{2}\sum_{i=1}^{N} \log|X_i^T V_i^{-1} X_i|,$$

for model with f fixed effects and N observations in total.

Random effects U_i are not included in the likelihood, therefore they can be predicted with V_i, X_i and Y_i given: $U_i = G_i Z_i^T V_i^{-1}(Y_i - X_i\gamma)$. Fixed effects γ and their SEs also can be estimated with V_i's given (similar to GLM estimation):

$$\gamma = \left(\sum_{i=1}^{N}(X_i^T V_i^{-1} X_i)\right)^{-1} \sum_{i=1}^{N}(X_i^T V_i^{-1} Y_i) \quad \text{and} \quad \mathbb{C}ov(\gamma) = \left(\sum_{i=1}^{N}(X_i^T V_i^{-1} X_i)\right)^{-1}$$

To fit a random linear time model we include `Age` variable as random factor in addition to the intercept:

```
model03 <- lmer(Weight ~ 1 + Age + (1 + Age|ChildID),
                    data=df_final, REML=TRUE)
plot_results(df_final, model03, n=5, ncols=5,
             model_name = "Model 3: random linear time (RLT)")
```

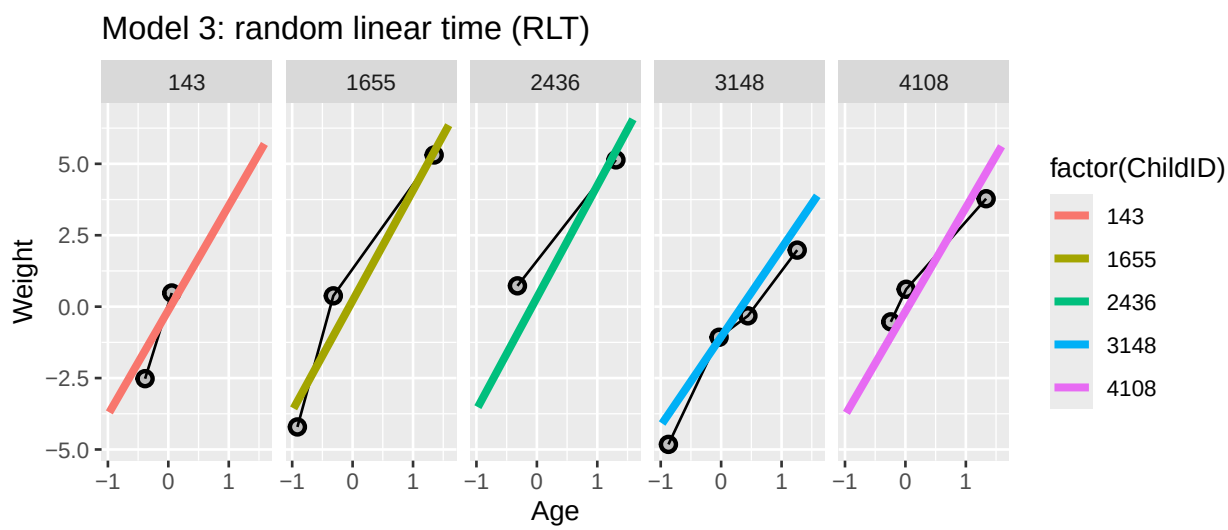

Figure 5: Model predictions with random intercept and random slope

Here we can see slight differences between the slopes for different subjects. However, the model fit is singular again, here both random effects are perfectly correlated. Still, we can compare three existing models using `anova()` function from `lmerTest` package for likelihood-ratio test. Results are collected in the Table 3, and we can see that inclusion of time variable and even its random effect increased the quality of the model.

```
anova(model01, model02, model03)
```

Table 3: Unconditional model comparison

Model	# par	AIC	BIC	-2*LL	χ^2_d	d	p-value
model01	3	7708.2	7724.1	7702.2			
model02	4	5357.5	5378.7	5349.5	2352.620	1	< 2.2e-16
model03	6	5286.2	5318.0	5274.2	75.362	2	< 2.2e-16

Conditional linear model

We now explored different assumptions about the time structure. The potential for further improvement remains, for example, through the inclusion of additional covariates. Here we have two time-invariant regressors - weight at birth as baseline observations (numeric) and gender(binary), - that can be included in the model. Theoretically, birth weight not only influences intercept, but also can have an impact on growth rate after birth. For example, children that are born premature, often have health issues directly after birth and may grow slower. It is also interesting to study, whether differences between development speed of boys and girls can be seen during early stages of life.

Since it is possible to test all hypotheses at once, we will include both fixed effects and also consider interactions between them and time variable. Then the equation on the level 1 will be written as follows:

$$y_{ti} = (\gamma_{00} + U_{0i}) + (\gamma_{10} + U_{1i}) \cdot x_{ti} + \gamma_{20} \cdot b_{ti} + \gamma_{30} \cdot g_{ti} + \gamma_{40} \cdot x_{ti} \cdot b_{ti} + \gamma_{50} \cdot x_{ti} \cdot g_{ti} + e_{ti}$$

where b_{ti} and g_{ti} denote birth weight and gender respectively. Design matrix X_i is updated accordingly by two columns and Z_i remains unchanged. To fit such model, we include respective variables into the formula and then review model summary (see Table 4).

```
model04 <- lmer(Weight ~ 1 + Age * Birthweight + Age * GenderID + (1|ChildID),
                data=df_final, REML=TRUE)
```

```
summary(model04)
anova(model02, model04)
```

Residuals are again distributed around zero. Random intercept U_{0i} has estimated variance of $\hat{\tau}^2_{U_0} = 0.25 \pm 0.50$, which has decreased compared to corresponding unconditional model (model02). Residual variance is estimated at $\hat{\sigma}^2_e = 1.76 \pm 1.33$ (same as in the second model). Fixed intercept estimate increased from -0.015 to 0.23, and fixed slope increased a little as well. Both new fixed effects seem to be significantly larger/smaller than 0 as well as interaction term between birth weight and age, however, absolute value of estimated parameters are rather small, except for fixed slope parameter. Also the test shows that the addition of these covariates significantly improves the fit (see Table 5).

Table 4: Conditional model parameter estimation

Predictor	Estimate	Std. Error	df	t value	p-value
(Intercept)	0.23682	0.05842	429.06364	4.054	5.98e-05
Age	3.71449	0.06264	1168.76560	59.295	< 2e-16
Birthweight	0.91200	0.08321	429.88095	10.960	< 2e-16
GenderID	-0.49817	0.08329	442.78809	-5.981	4.58e-09
Age × Birthweight	0.18107	0.08929	1183.33744	2.028	0.0428
Age × GenderID	-0.09151	0.08915	1172.60511	-1.026	0.3049

Table 5: Conditional vs. unconditional model comparison

Model	# par	AIC	BIC	-2*LL	χ^2_d	d	p-value
model02	4	5357.5	5378.7	5349.5			
model04	8	5219.0	5261.4	5203.0	146.59	4	< 2.2e-16

Beyond linearity: quadratic time effect

However, there is still some possibility for improvement. During data exploration we mentioned that the growth doesn't seem to be linear: weight increases very quickly short after birth but then the growth slows down. We can try to account for that in the model by including non-linear transformation of the time variable into the models for mean and/or for variance.

```
model05 <- lmer(Weight ~ 1 + Age + I(Age^2) + (1|ChildID),
                data=df_final, REML=TRUE)
model06 <- lmer(Weight ~ 1 + Age + I(Age^2) + (1 + Age|ChildID),
                    data=df_final, REML=TRUE)
plot_results(df_final, model06, n=5, ncols=5,
             model_name = paste(c("Model 6: random intercept, random linear,",
                                  "fixed quadratic time")))
```

```
anova(model02, model05, model06)
```

When extending the model to account for non-linear development, a quadratic function of time yielded a more accurate representation of the data (compare Table 6). In this case, models incorporating both fixed and random linear time effects were appropriate, however, quadratic term of time should be rather included as fixed effect only. This indicates that the observed trajectories are not purely linear but exhibit curvature over time. However, quadratic function of time cannot be an optimal solution for long-term prediction. By definition it starts to decrease after some time with the same rate it was increasing before, which cannot be an adequate model for body weight.

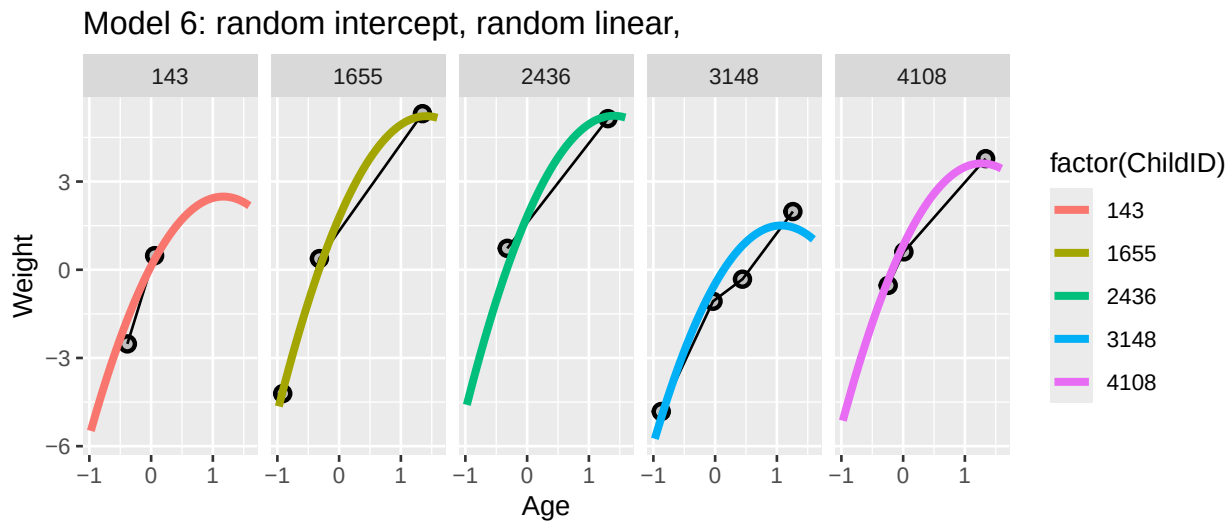

Figure 6: Model predictions with quadratic time

Table 6: Model comparison for linear vs. quadratic time

Model	# par	AIC	BIC	-2*LL	χ^2_d	d	p-value
model02	4	5357.5	5378.7	5349.5			
model05	5	4356.0	4382.5	4346.0	1003.54	1	$<$ 2.2e-16
model06	7	4080.5	4117.6	4066.5	279.49	2	$<$ 2.2e-16

Summary and outlook

Analyzing longitudinal data with multilevel models reveals some key practical and methodological aspects. First key observation concerns the robustness of multilevel models in the presence of missing and imbalanced data. Longitudinal datasets are often characterized by unequal numbers of observations across individuals or irregular measurement intervals. The applied modeling framework proved capable of handling such cases effectively, which represents a substantial advantage over more traditional approaches such as repeated-measures ANOVA. Furthermore, the results emphasize the crucial role of data pre-processing. Proper handling of missing values and scaling of variables are essential not only for ensuring model convergence but also for enabling meaningful interpretation of the estimated parameters.

Several directions for future research can be formulated based on the results of this analysis. Firstly, other structures for variance models can be explored. Here, only unstructured variance model was used, but it is possible that other structures such as compound symmetry (CS) or autoregressive structure will describe the variability in the observations better. Secondly, alternative non-linear transformations of time variable can be explored. For example, logarithmic or exponential functions may provide a better fit for certain types of growth or decay processes in long term studies, particularly when changes occur more rapidly at early time points and stabilize later on. The flexibility of multilevel models allows such transfor-

mations to be incorporated straightforwardly. Model comparison techniques and diagnostic tools should be employed to assess whether such extensions are justified by the data.

In conclusion, multilevel modeling provides a powerful and flexible framework for analyzing longitudinal data, enabling researchers to capture both overall trends and individual trajectories. Future developments and extensions of this approach hold considerable potential for advancing empirical research in this field.

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